DEEPAK SINGH

AI / Machine Learning Intern

Linkedin | Github | Portfolio | Gmail

PROFESSIONAL SUMMARY

B.Tech CSE (2026) graduate and AWS-certified AI engineer with certifications in **AWS Certified Cloud Practitioner** and **AWS Certified AI Practitioner**. Hands-on experience in LLM fine-tuning, prompt engineering, RAG pipelines, and deploying ML systems using FastAPI, Docker, and CI/CD.

Proficient in Python, machine learning, deep learning, and MLOps, with experience building end-to-end AI pipelines, scalable backend APIs, and cloud-based applications using AWS and GCP. Focused on developing and deploying production-ready AI solutions.

TECHNICAL SKILLS

Programming: Python, SQL, Java

Machine Learning: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Feature Engineering, Model Evaluation, Hyperparameter Tuning

LLMs & NLP: LLM Fine-Tuning (LoRA / PEFT), Hugging Face Transformers, Prompt Engineering, Structured Information Extraction, Text Classification, RAG, FAISS, Agentic Workflows

Deployment & MLOps: FastAPI, REST APIs, Docker, Jenkins (CI/CD), MLflow, GCP, AWS, Git, n8n

EXPERIENCE

AI Engineer Intern (CyCrew) Oct 2025 – Jan 2026

- Built an LLM- and RAG-based incident analysis platform that turns raw security alert data into **structured, LLM-generated reports**, automating alert investigation for Security Operations Center (SOC) teams.
- Designed prompt engineering and retrieval strategies (FAISS, embeddings) to ground LLM outputs in alert context, improving accuracy and consistency.
- Developed an end-to-end AI pipeline with a FastAPI backend, n8n workflow orchestration, and a Vue.js frontend, **deploying the model as a served API** that reduced manual triage effort.

PROJECTS

Legal Document Classification (LoRA Fine-Tuning of Legal-BERT) Kaggle Competition

- Fine-tuned **Legal-BERT** with **LoRA (PEFT)** in PyTorch and Hugging Face to classify ~50K long-form EU legal documents into 30 classes, training only ~0.8% of model parameters and achieving **85% validation accuracy**.

Hotel Reservation Prediction (Booking Cancellation Risk) GitHub

- Developed a production-ready ML pipeline to predict booking cancellations, a risk-classification problem, covering data ingestion, cleaning, feature engineering, model training, and evaluation.
- Implemented GCP-based data ingestion, hyperparameter tuning, and MLflow experiment tracking; containerized with Docker and automated deployment through Jenkins CI/CD.

Virtual Data Analyst (LLM-Powered AI Agent) GitHub

- Built an API-based AI agent that sources, preprocesses, analyzes, and visualizes datasets end-to-end with minimal user input.
- Used LLMs for dynamic metadata extraction, letting the agent adapt its cleaning and analysis workflow to each dataset's structure.

EDUCATION

B.Tech CSE (Dronacharya College of Engineering) Jun 2022 – Jul 2026

BS in Data Science and Applications (IIT Madras) Sep 2022 – March 2027

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services
- AWS Certified AI Practitioner — Amazon Web Services
- Deep Learning for Natural Language Processing — NPTEL (Top 5%)
- Amazon ML Summer School 2024 — Amazon
- Microsoft Applied Skills: Build a Natural Language Processing Solution with Azure AI Language — Microsoft
- FastAPI — Udemy (Issued Jun 2026)